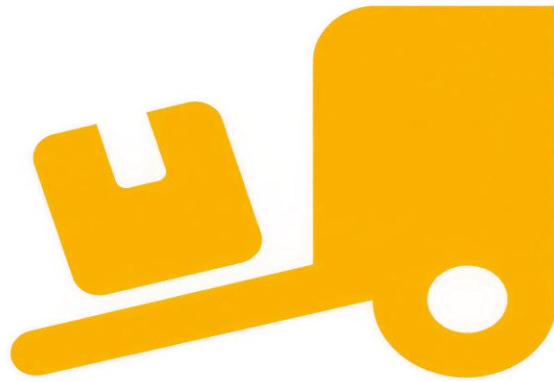




LogiTrack

Logistics & Delivery Management System

INTRODUCTION

Software testing is the process of checking a software system to find errors and to make sure that it works according to the required specifications. Testing verifies whether the system gives the expected output for different inputs and conditions.

The LogiTrack system is tested to ensure that its major functions work correctly and that users can complete the required logistics operations smoothly. The testing covers the main user functions, shipment processing, payment, delivery, tracking, database operations and reports.

TESTING OBJECTIVE

- To identify errors and defects in the system.
- To verify that all major modules work according to the requirements.
- To check whether valid and invalid inputs are handled correctly.
- To verify that customer, driver, branch manager and administrator operations work properly.
- To verify correct storage and retrieval of data.
- To check shipment booking, cost calculation and tracking.
- To verify advance payment and balance-payment operations.
- To verify delivery, proof of delivery and invoice generation.
- To verify container and shipment-transfer operations.
- To verify report generation and export.
- To ensure the system is reliable and user friendly.

TESTING METHODS

1. Black Box Testing

Black box testing checks the system from the user's point of view without considering the internal program code. Inputs are given and the actual output is compared with the expected output.

2. White Box Testing

White box testing checks the internal logic, program conditions, database operations and functions of the system.

TESTING STEPS

1 Unit Testing

Each important function is tested separately to make sure that it produces the expected result.

Examples: login validation, password verification, shipping-cost calculation, shipment creation, payment processing, invoice generation and report generation.

2 Integration Testing

Integration testing checks whether different modules work together correctly. For example, a shipment booked by the customer should be stored in the database, payment should be recorded, and the shipment status should be updated correctly.

3 Validation Testing

Validation testing checks whether the completed system satisfies the requirements of its users. Customer, driver, branch manager and administrator functions are checked.

4 Output Testing

Output testing checks whether the displayed information and generated documents are correct and shown in the required format. This includes shipment details, payment status, tracking information, invoices and reports.

User Acceptance Testing

User acceptance testing checks whether the system is convenient, understandable and suitable for its intended users.

TEST CASES

The following are the main test cases performed for the LogiTrack system.

Module	Test Condition	Expected Result	Result
User Login	Enter valid login details	User is logged in successfully	Pass
User Login	Enter invalid login details	Error message is displayed	Pass
Registration	Enter valid registration details	New account is created	Pass
Registration	Leave required field empty	Validation message is displayed	Pass
Shipment Booking	Enter valid shipment details	Shipment is created successfully	Pass
Cost Calculation	Enter valid weight and shipment options	Correct shipping cost is calculated	Pass
Tracking	Enter a valid tracking number	Correct shipment status is displayed	Pass
Tracking	Enter an invalid tracking number	Shipment-not-found message is displayed	Pass
Advance Payment	Make valid advance payment	Payment is recorded and shipment status is updated	Pass
Pickup	Driver enters actual package weight	Weight and shipping cost are updated correctly	Pass
Balance Payment	Make remaining payment during pickup	Payment is recorded successfully	Pass
Cash Handover	Driver submits collected cash	Cash status is updated	Pass
Invoice	Generate shipment invoice	Invoice is generated successfully	Pass
Container	Create and transfer shipment container	Container and shipment status are updated correctly	Pass
Delivery	Complete delivery without required proof	System prevents final delivery	Pass

Delivery	Complete delivery with photo and signature	Shipment is marked as delivered	Pass
Reports	Open required report	Correct report information is displayed	Pass
Reports	Export report	Requested report file is generated	Pass
Database	Create/update shipment information	Correct data is stored and retrieved	Pass
Role Access	Open module using authorized role	Correct module is accessible	Pass

1 LOGIN AND REGISTRATION TESTING

Test Condition	Expected Result	Result
Valid credentials are entered	User is logged in successfully	Pass
Invalid credentials are entered	Login error is displayed	Pass
Valid registration details are entered	Account is created	Pass
Required field is empty	Validation message is displayed	Pass
Invalid details are entered	System rejects invalid input	Pass

2 SHIPMENT AND PAYMENT TESTING

Test Condition	Expected Result	Result
Valid shipment details are entered	Shipment is created	Pass
Shipment cost is calculated	Correct cost is displayed	Pass
Advance payment is completed	Payment is recorded and shipment is confirmed	Pass
Package is weighed at pickup	Actual weight and cost are updated	Pass
Balance payment is completed	Remaining amount is recorded	Pass

3 DELIVERY AND TRACKING TESTING

Test Condition	Expected Result	Result
Valid tracking number is entered	Shipment tracking details are displayed	Pass
Invalid tracking number is entered	Error message is displayed	Pass
Delivery is attempted without proof	System prevents completion	Pass
Photo and receiver signature are provided	Shipment is marked Delivered	Pass
Invoice is requested	Invoice is generated successfully	Pass

4 DATABASE TESTING

Test Condition	Expected Result	Result
User data is entered	Data is stored correctly	Pass
Shipment is created	Shipment data is stored correctly	Pass
Payment is completed	Payment data is stored correctly	Pass
Shipment is updated	Updated values are retrieved correctly	Pass
Report data is requested	Correct records are retrieved	Pass

5 REPORT TESTING

Test Condition	Expected Result	Result
Shipment report is opened	Shipment information is displayed	Pass
Revenue report is opened	Revenue information is displayed	Pass
Driver/customer report is opened	Required details are displayed	Pass
Branch/vehicle report is opened	Required details are displayed	Pass
Report is exported	File is generated successfully	Pass

TESTING SUMMARY

The LogiTrack system was tested using different testing methods to verify its major functionalities. The testing covered login and registration, shipment booking, shipping-cost calculation, payment, pickup, tracking, container movement, delivery, proof of delivery, invoice generation, database operations and reports.

The test cases were designed using valid and invalid inputs so that both normal and abnormal conditions could be checked. The expected results were compared with the actual system behaviour. The major functions were found to work as expected.